MARIUS EMANUEL HEES

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Erasmus School of Economics, Erasmus University Rotterdam
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Up to date: September 2026

PLACEMENT: Tinbergen Institute
Prof. Eric Bartelsman, placement director
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Christina Månsson, placement assistant
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MAIN INTERESTS

I am an economist working on urban, spatial, environmental, and development economics. My research studies how space and networks shape economic activity, and how these dynamics, through migration, trade, infrastructure, and climate adaptation, reshape economic outcomes. Combining structural modeling, field experiments, and spatial methods with satellite and micro data, I aim to address key challenges in climate risk and resilience, migration and urbanization, urban labor markets and informality, transport and infrastructure, trade networks, and structural transformation. I am on the 2026–2027 job market.

FIELDS OF CONCENTRATION

PRIMARY
Development Economics
Urban, Spatial and Trade Economics
Environmental and Climate Economics

SECONDARY
Applied Econometrics
Migration and Labor Economics
Quantitative Spatial Economics

DISSERTATION TITLE

"Climate, Migration and Structural Transformation: Satellite Data, Field Experiments and General Equilibrium Models for Resilient Development"

Expected Date of Completion: Spring 2027
Principal Advisors: Professor Frank van Oort and Dr. Michiel Gerritse

PAPERS

JOB MARKET PAPER

"Segmented Labor Markets and Structural Transformation"

Abstract: Most of this century's urban growth will occur in low-income economies. Whether urban growth turns into economic growth depends on the capacity of cities to absorb entrants into productive employment. Most migrants first enter informal work, and long waits for formal jobs can sever the link between urbanization and structural transformation. To formalize this tension, I develop a dynamic spatial general-equilibrium model with separate informal and formal urban labor markets and congestible access to formal employment. Migration fills an informal queue, larger queues slow formalization, and weaker prospects feed back into migration. Quantified for Ethiopia using newly assembled satellite and survey microdata, the model shows that current formalization capacity is too weak to sustain structural transformation. Without reform, the economy settles into an informality trap in which cities keep growing but remain largely informal. Closing this capacity gap raises long-run GDP by 13.1 percent and national welfare over the transition by 9.7 percent. Formalization capacity shapes what policies achieve and who benefits. Policies that raise productivity or purchasing power in rural and informal employment remain effective when formalization is slow, while mobility and formal-sector interventions generate broader gains as the queue clears. These policy effects can run in opposite directions across workers within the same city, a pattern a one-market city cannot represent. The same margin determines whether workers displaced by climate change and conflict move into productive employment or accumulate in informal congestion. Cities deliver growth not when workers arrive, but when labor markets absorb them into productive employment.

Presented at: North American Meeting of the Urban Economics Association (Chicago, 2026); Dutch Urban Economics and Trade Workshop (2025, 2026).

WORKING PAPERS

“(Mis)measuring Local Weather Shocks: Evidence from Drought and Migration in Ethiopia”
(with Michiel Gerritse and Frank van Oort). Under review.

Abstract: Extreme weather shocks strike locally, yet their effects are typically estimated from exposure aggregated across districts, regions or countries. We show that this aggregation systematically biases impact estimates and propose a simple correction. Using high-resolution rainfall and satellite-derived population data for more than 13,000 Ethiopian settlements (2000–2016), we find that drought reduces population in vulnerable settlements by about 2.5 percent, concentrated among working-age adults and coinciding with falling food production, income and health. Measured at progressively coarser scales, the impact disappears, even at resolutions common in the literature. We decompose the bias into three sources (partial exposure misclassified, local vulnerability averaged away, area weights misrepresenting where people live) and derive a fix: measure each shock at fine scale, weight by local vulnerability, then aggregate. The correction recovers estimates in areas roughly four times larger than those at which the standard binary indicator fails; simulations confirm its performance. Building exposure locally before aggregating is a low-cost change that can substantially improve empirical practice.

Presented at: 6th GCD Workshop on Regional Development and Innovation (A Coruña, 2024); Regional Science Workshop on Remote-Sensed Identification and Networks (Oviedo, 2024); PSE Summer School on Migration (2023); ESE/TI internal seminars.

“Memory, Expectations, and Climate Adaptation”
(with Frank van Oort). Draft, September 2026. Pre-registered field experiment (AEA RCT Registry, 2026).

Abstract: Smallholder farmers face increasingly frequent weather shocks, yet take-up of insurance and other protection remains low. We show that part of this demand problem lies in how farmers form beliefs about future weather. In a survey experiment with 686 farmers in eastern Uganda, we elicit memories of past seasons and expectations for the coming one, benchmark them against remotely sensed weather records, and experimentally provide information on local weather history. Memory is strongly compressed—large differences in the record become small differences in recall—and this compression carries into expectations, while memorable extreme events form a separate, less compressed channel. Beliefs matter: farmers expecting adverse weather plan more adaptation. A simple visual summary of local weather history moves expectations toward the record, most for those furthest away; literacy predicts accuracy but not responsiveness. Belief formation is thus a demand-side margin in climate adaptation, and accessible information a low-cost complement to protection policies.

Presented at: ESE/TI internal seminars.

EDUCATION

	LEVEL	DATE	FIELD
Erasmus University Rotterdam & Tinbergen Institute	PhD	2027 (exp.)	Economics
University of Chicago (Host: E. Rossi-Hansberg)	Visiting PhD	2024	Economics
University of Oxford	MPhil	2022	Economics
Erasmus School of Economics	BSc	2019	Economics
Harvard University	Visiting BSc	2018	Economics

PROFESSIONAL EXPERIENCE

Research Assistant to Dr. Niclas Moneke, University of Oxford — 2021
Spatial development project on mobile-phone network expansion and economic development

Research Assistant, ifo Institute for Economic Research, Munich — 2020
Project with Prof. Niklas Potrafke on trade networks (Center for Public Finance and Political Economy)

Research Assistant and Consultant, United Nations ESCAP, Bangkok — 2019
Cross-national rail connectivity across the Asian continent (Transport Connectivity and Logistics Section)

Research Assistant, University of Tehran — 2019
Data collection on local labor market dynamics

TEACHING

Lecturer, Urban Economics (MSc), Erasmus School of Economics — 2025

Foundational course of the MSc programme in Urban, Port and Transport Economics. Designed and taught the module on developing-country cities: lecture, R-based problem set and seminar on general-equilibrium analysis of urban transport policy, and assessment

Lecturer, Firm Location Choice and Productivity in Cities (BSc), Erasmus School of Economics — 2022–2025

BSc Strategy Economics seminar, four cohorts: weekly session reports, literature surveys, student paper presentations and quizzes; grading and feedback for 100–120 students per cohort

Thesis Supervision, Erasmus School of Economics — 2022–2026

BSc theses in Strategy Economics (15) and Urban, Port and Transport Economics (3); MSc theses in Strategy Economics (19) and Urban, Port and Transport Economics (4). Topics include optimal transport-network allocation, carbon pricing and local pollution, energy-crisis resilience of firms, and urban regeneration.

SKILLS

Programming	R, Julia, Python, Stata, MATLAB, Bash, Git, LaTeX
Spatial and remote sensing	ArcGIS, QGIS, Google Earth Engine; large-scale satellite-data pipelines
High-performance computing	Snellius national supercomputer (SURF); parallel and batch workflows
Languages	German (native), English (fluent), French (C1), Dutch (A2)

WORKSHOPS, SEMINARS AND CONFERENCES

North American Meeting of the Urban Economics Association, Chicago	2026
Dutch Urban Economics and Trade Workshop, Groningen	2026
Dutch Urban Economics and Trade Workshop, Rotterdam	2025
6th GCD Workshop on Regional Development and Innovation, A Coruña	2024
Regional Science Workshop on Remote-Sensed Identification and Networks, Oviedo	2024
Summer School on Migration (Giovanni Peri), Paris School of Economics	2023
ESE and Tinbergen Institute internal seminars	2023–2026

FELLOWSHIPS, GRANTS AND AWARDS

Erasmus Research Institute of Management (ERIM) — €11,000 — 2025–2026

Field Experiment in Uganda: PhD Support and Data Collection Grants

Tinbergen Institute, ESE and Erasmus Professor funds — €16,000 — 2025–2026

Field Experiment in Uganda: Internal research funding

Erasmus Trustfonds — €5,000 — 2024

Research visit, University of Chicago

German Academic Scholarship Foundation (Studienstiftung) — €40,000 — 2020–2022

MPhil scholarship

German Academic Scholarship Foundation (Studienstiftung) — €45,000 — 2016–2019

BSc scholarship

OTHER INFORMATION

Citizenship: Germany and France (EU); United Kingdom pre-settled status

REFERENCES AND DISSERTATION COMMITTEE

Prof. Frank van Oort (advisor)
Erasmus School of Economics
Erasmus University Rotterdam
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Dr. Michiel Gerritse (advisor)
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Prof. Esteban Rossi-Hansberg
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